Remzi Efe Demir

Mechatronics Engineering Student

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EDUCATION

Atılım University, *B.Sc. Mechatronics Engineering* 2023 - 2028 (expected)

EXPERIENCE

Neutronix Technologies, *Voluntary Intern* Aug 2026 - Present

- Contributed to preparing Lastenheft, SOP and test report documents.
- Designed schematics and PCBs in Altium Designer and ran circuit simulations in LTspice.

Project: 40 W Flyback Power Supply

- Handled the converter design, simulation and PCB layout from start to finish.
- Prototype tests for efficiency, load regulation, output ripple and thermal performance are planned.

MZD Elektrik Elektronik A.Ş., *Mechatronics Engineering Intern* Jul 2026 - Aug 2026

- Rotated through 4 departments of a defense cable-harness manufacturer and ROKETSAN strategic partner.
- Followed the production planning process in IFS ERP from work order to shipment.

QuadRobotics R&D and Software, *Voluntary Intern* Jan 2025 - Mar 2025

- Provided SolidWorks design support for payload integration on the ROKETSAN KOZ quadruped robot platform.
- Learned the MIL-STD-810 and MIL-STD-498 standards and the design processes that comply with them.
- Worked on payload hardware integration for robot dog platforms and NACA-based airfoil design.

PROJECTS

High-Autonomy Reconnaissance Drone, *Project Leader*, *4-person team* 2026 - Present

- Project approved by HAVELSAN and AFAD; development is ongoing.

Autonomous Tracking and Moving-Platform Landing Drone, *Project Leader*, *6-person team* 2026 - Present

- Developing a rotary-wing UAV that autonomously tracks a target and takes off from and lands on a moving platform.

Swarm UAV for GPS-Denied Environments, *Team Captain*, *17-person team* Nov 2024 - Aug 2026

- Developed a low-cost drone swarm capable of formation flight in GPS-denied environments.
- Passed the Preliminary Design Report and video stages of the TEKNOFEST Swarm UAV Competition; the project was supported by the Atılım University Pre-Incubation Center.
- Designed the airframe in SolidWorks and produced it by 3D printing; designed the flight controller carrier board in Altium and developed an original PID flight control algorithm on ESP32.
- The project was accepted into the TÜBİTAK 2209-B Industry-Oriented Undergraduate Research Projects program; TÜBİTAK work is ongoing.

Autonomous Underwater Vehicle (AUV), *Electronics Team Leader* Sep 2024 - Aug 2026

- Joined as a team member and became electronics team leader within 3 months.
- Designed the power electronics and battery system; led PCB design, sensor integration and communication infrastructure.
- Modeled an underwater environment in Gazebo with ROS 2 Humble; integrated OpenCV and YOLO-based vision nodes to achieve autonomous motion of the vehicle in simulation.
- Co-authored the MÜBAK'25 paper; presented the vehicle at MÜBAK'26.

Unmanned Surface Vehicle (USV), *Electronics Team Leader* 2024 - 2025

- Designed the electronic system architecture in Altium Designer and set up the RF link to the kamikaze drones carried on board; presented the project at MÜBAK'25.

Wearable Health Tracking Device, *Embedded Software* 2024 - 2025

- Developed software and performed calibration for sensors communicating over I2C, SPI and UART; the project won 1st prize at Ankara University.

SKILLS

Electronics: Altium Designer, PCB design, power electronics (flyback), LTspice, BMS, RF communication, I2C / SPI / UART
Embedded: STM32, ESP32, Arduino, Raspberry Pi, NVIDIA Jetson, C/C++
Software & Control: Python, MATLAB, ROS 2 (Humble), Gazebo, Linux, PID control, Kalman filter, OpenCV, YOLO
Mechanical: SolidWorks, Siemens NX, 3D printing
Standards & Tools: MIL-STD-810, MIL-STD-498, SOP, Lastenheft, IFS ERP
Languages: Turkish (native), English (B2), Danish (basic)